

Unit 3, Lesson 1: Rearranging Equations of Lines



Benchmarks

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.F.1.5 Compare key features of linear functions each represented algebraically, graphically, in tables, or written descriptions.



Learning Targets

- ☐ I can rearrange the equations of a line from one form into another form.
- ☐ I can relate key features of lines to different forms of linear equations.



3.1.1 Warm-Up: Solving Equations

1. Solve each equation by isolating the variable. Show your work.

a. $14 = 3x - 28$

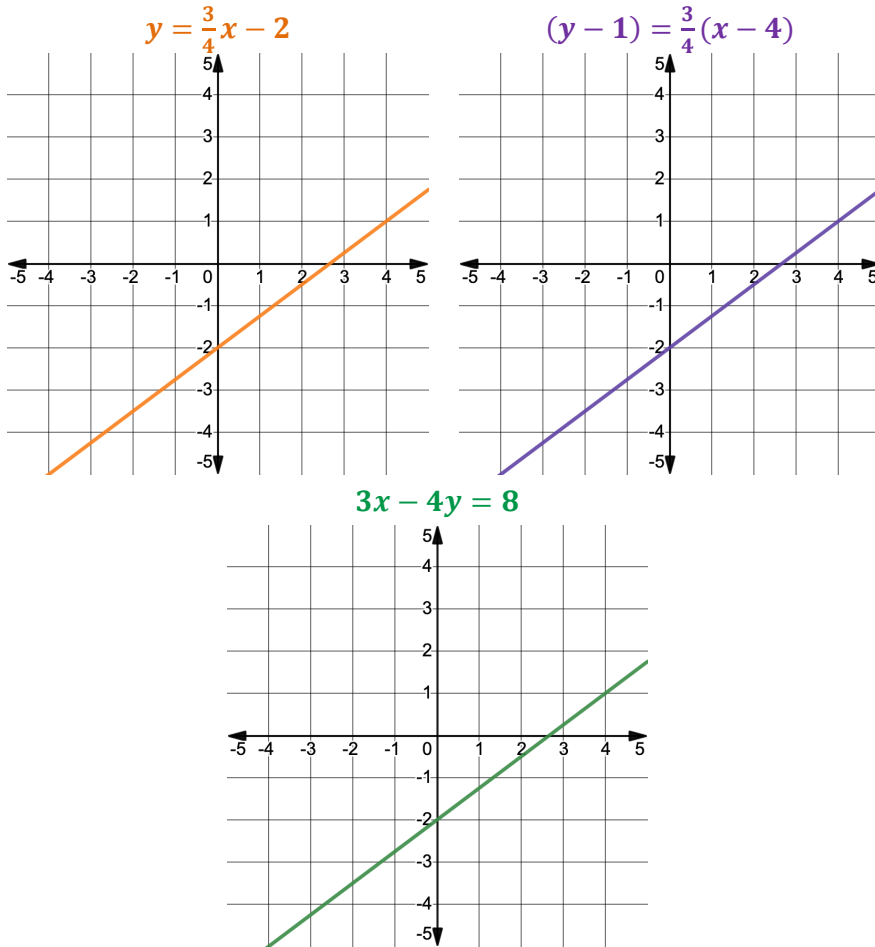
b. $x + 19 = 6(x - 11)$

c. $27 + 9y = -12$



3.1.2 Guided Instruction: Different Forms of Linear Equations

Three linear equations and their graphs are shown.



1. Discuss with your partner what you notice about the graphs and equations. Summarize your discussion.

The equations of lines can be written in different but equivalent forms. Each form of the equation relates to different features on the graph of the relationship, including the slope, y-intercept, and other points on the line.



The **slope** of a line is the ratio of the change in the vertical direction to change in the horizontal direction, often expressed as the $\frac{\text{change in } y}{\text{change in } x}$, or $\frac{\Delta y}{\Delta x}$.



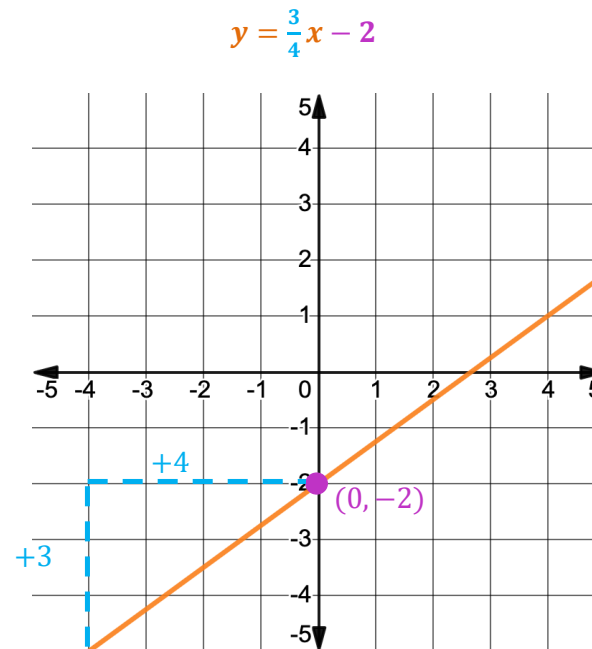
The **y-intercept** is the value of y at the point where a line or graph intersects the y -axis. The value of x is 0 at this point.

One form for the equation of a line is slope-intercept form.

Slope-Intercept Form

$$y = mx + b$$

where m is the slope and b is the y-intercept



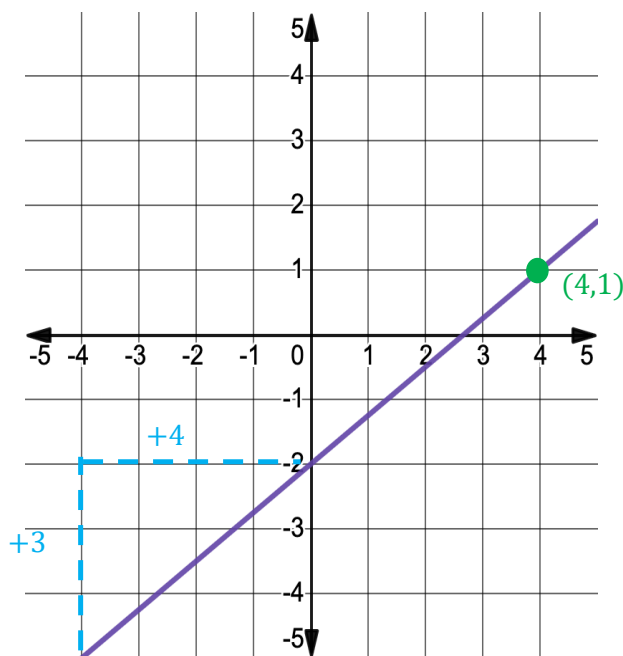
2. Discuss with your partner why this form is called slope-intercept form. Summarize your discussion.

A second form for the equation of a line is called point-slope form.

Point-Slope Form

$(y - y_1) = m(x - x_1)$
where m is the slope and (x_1, y_1) is any point on the line

$$(y - 1) = \frac{3}{4}(x - 4)$$



Point-slope form can be written using the slope and any point on the line.



3. Compare the equation and the graph of the line. What do you notice about the signs of the coordinates of the green point used to write the equation compared to the values in the equation?

4. Katrina wrote a different equation for the same line in point-slope form using the y -intercept, $(0, -2)$. Her equation is shown.

$$(y + 2) = \frac{3}{4}(x - 0)$$

- a. Explain why the expression $(y + 2)$ is on the left side of the equation when the y -value of the point used is -2 .
- b. Discuss with your partner how the equation $(y + 2) = \frac{3}{4}(x - 0)$ could be rewritten in slope-intercept form.

- c. Rewrite the equation $(y + 2) = \frac{3}{4}(x - 0)$ in slope-intercept form by isolating the variable, y .

- d. Rewrite the original point-slope equation, $(y - 1) = \frac{3}{4}(x - 4)$, in slope-intercept form.

- e. What do you notice about the equations generated in parts c and d?

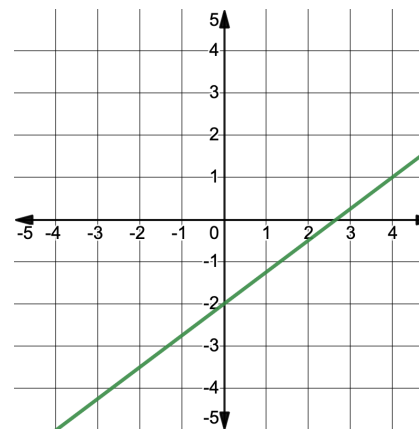
The third form for the equation of a line is called standard form.

Standard Form

$$Ax + By = C$$

where A , B , and C are rational numbers and A and B cannot both be zero

$$3x - 4y = 8$$



5. Rewrite the equation $3x - 4y = 8$ in slope-intercept form by isolating the variable, y .



A **linear expression** or **linear equation** is a polynomial expression or equation that contains a term of degree 1, but no term of higher degree.



3.1.3 Guided Practice: Rewriting Equations of Lines in Different Forms

- Use the equation $f(x) = \frac{2}{5}x - 4$ to complete the following.
 - What is the form of the linear equation given?
 - Rewrite the equation in standard form.
- Use the equation $(y - 7) = -\frac{1}{3}(x + 2)$ to complete the following.
 - What is the form of the linear equation given?
 - Rewrite the equation in standard form.
- Use the equation $6x - 9y = -12$ to complete the following.
 - What is the form of the linear equation given?
 - Rewrite the equation in slope-intercept form.



3.1.4 Collaboration: Equivalent Equations

Twelve linear equations are given in various forms.

$4x - 6y = 24$	$-4x - 6y = 24$	$-6x + 4y = 24$	$6x + 4y = 24$
$y = -\frac{2}{3}x - 4$	$f(x) = -\frac{2}{3}x - 6$	$y = \frac{3}{2}x + 6$	$f(x) = \frac{2}{3}x - 4$
$(y + 6) = -\frac{3}{2}(x - 8)$		$(y - 6) = \frac{3}{2}x$	
$(y + 6) = \frac{2}{3}(x + 3)$		$(y - 6) = -\frac{2}{3}(x + 15)$	

- Using the equations, work with your partner to complete the table so that each row contains a set of equivalent equations, one written in each form. Not all equations will be used.

Equivalent Equations	Standard Form	Slope-Intercept Form	Point-Slope Form
Set 1			
Set 2			
Set 3			



3.1.5 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. Today I am still unsure about ...

2. I will try to build my confidence by ...

Unit 3, Lesson 2: Writing Equations of Lines



Benchmarks

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Targets

- ☐ I can determine the slope of a line from its graph.
- ☐ I can write the equation of a line given its graph.
- ☐ I can rearrange the equations of a line from one form into another form.